Thesis VT26B

Submission deadline in 7.3 hours · abhishekyadav.info1@gmail.com ☰

#41 Uncertainty-Aware Machine Learning for Microwave-Based Body Composition Assessment

Main Edit Your submissions

Updated submission (changed Research)

This submission is marked as not ready for review. You have until Tuesday Jan 6, 2026, 10:59 PM UTC (11:59 PM your time) to make further changes. Submissions incomplete as of Tuesday Jan 6, 2026, 10:59 PM UTC (11:59 PM your time) will not be considered.

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* Required

Title *

Uncertainty-Aware Machine Learning for Microwave-Based Body Composition Assessment

Submission (PDF, max 600MB) *

abhishek_yadav_thesis_proposal-signed_final.pdf (304kB) Jan 6, 2026, 3:36:03 PM UTC · ada3c8ca

Replace

Abstract *

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Noninvasive microwave sensing has emerged as a promising technique for biomedical diagnostics. However, a critical limitation for clinical translation is that current machine learning models typically output point estimates without prediction confidence. In real clinical settings, measurement noise and biological variability can lead to unreliable predictions. This thesis aims to develop and evaluate an uncertainty-aware machine learning system for estimating skin, fat, and muscle properties. By integrating Bayesian techniques and quantifying predictive uncertainty, this work seeks to provide a clinically interpretable risk score to support safer medical decision-making.

Authors *

List the authors, including email addresses and affiliations.

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Contacts

These users can view and edit the submission. All listed authors with site accounts are contacts. You can add contacts who aren't in the author list or create accounts for authors who haven't yet signed in.

☑ Abhishek Yadav <abhishekyadav.info1@gmail.com>

Add contact

Personnummer *

In this format (ÅÅÅÅMMDD-XXXX). This will be used to check your records in Ladok. Make sure it is correct especially if you have had a temporary one before. Otherwise this will cause delays in your application.

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20000715-8918

Program *

Select the program you belong to.

- ○ Computer Science Bachelor
- ◉ Computer Science Master
- ○ IT Engineering
- ○ Computational Science
- ○ Embedded Systems
- ○ Image Analysis and Machine Learning
- ○ Data Science
- ○ Industrial Analytics
- ○ Sociotechnical Systems Engineering
- ○ Physics
- ○ Electrical Engineering
- ○ Mathematics
- ○ Industrial Engineering and Management
- ○ Molecular Biotechnology Engineering
- ○ Engineering Physics
- ○ Materials Engineering
- ○ Other

Program (if other)

If your program is not on the above list, type it here. It should match the description in Ladok and in English.

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Credits *

Number of credits.

- ○ 15
- ◉ 30

Location *

Internal refers to projects taking place in the IT Department. External refers to projects taking place in another department, a company or organisation.

- ◉ Internal
- ○ External

Company/Department *

The company name if external, the department name if Uppsala University.

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Information Technology

Supervisor name *

If there is more than one supervisor, please separate the names with a comma. Do not add the supervisors name in the author list or the contacts list.

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Bappaditya Mandal

Supervisor email *

If there is more than one supervisor, please separate the emails with a comma. Do not add the supervisors email in the author list or the contacts list.

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bappaditya.mandal@angstrom.uu.se

Research *

The IT department conducts research in the following areas. Pick an area that aligns most with your project.

- ○ Human-Machine Interaction
- ○ Computational Science
- ○ Image Analysis
- ○ Computer Systems
- ○ Optimisation
- ○ Computing Education Research
- ○ Parallel and Distributed systems
- ○ Control and Dynamical systems
- ○ Programming Languages and Systems
- ○ Cybersecurity
- ○ Semantics and Verifications
- ◉ Data Science
- ○ Software engineering

- ○ Embedded and Real-Time Systems
- ○ Artificial intelligence

Keywords *

Use 3 keywords to describe your project separated by a semicolon (e.g. MRI scans; Image segmentation; Supervised ML).

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Microwave Sensing; Uncertainty Quantification; Machine Learning

Confidentiality agreement *

Does your project require a confidentiality agreement? You can read more about it in the guidelines document here.

- ○ Yes
- ◉ No

Confidentiality agreement signed (PDF, max 600MB)

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Period *

Thesis start time.

- ◉ Period 3
- ○ Period 4

New or returning student *

New student refers to a student taking the course for the first time. Returning student refers to a student who discontinued the course or left the course abruptly.
Please note that you will have to contact the examiner to be reregistered to the course. Read more about it here under *Q. What do I need to do if I am a returning student who would like to continue the course?*

- ◉ New
- ○ Returning

Applied to thesis course *

Via www.universityadmissions.se (or antagning.se).

- ◉ Yes
- ○ No

Additional Information

For example, have you asked for the examiner's permission for reregistration if you are a returning student? Did you take an official leave of absense if you are a returning student?
This information will be visible to all potential subject reviewers and will be used to facilitate the matching between your project and the subject. reviewer. So use it to your advantage.

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Additional documents (PDF, max 600MB)

For example, evidence of approval for reregistration from the examiner, an appendix, etc..
Submit all the additional documents in one pdf.

Upload

Reviewer *

If, due to a special arrangement, you already have a reviewer, add their name here. The project plan should also contain the signature of the reviewer. In the absence of a signature, the answer is no. Do not add the reviewers name in the author list.

- (●) Yes
- ( ) No

Reviewer email

Do not add the reviewers email in the author list or the contacts list.

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Robin.Augustine@angstrom.uu.se

[x] **Confirm** *

I confirm that all information provided in this form is accurate and complete. I understand that providing false or misleading information will result in the cancellation of my application.

[ ] **The submission is ready for review**

ⓘ You can update this submission until Tuesday Jan 6, 2026, 10:59 PM UTC (11:59 PM your time). Only submissions marked ready for review will be evaluated.

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